

Cybersecurity Risk Assessment and Risk Register

1. Project description

In this project, I conducted a cybersecurity risk assessment for a commercial banking environment. I evaluated potential threats affecting critical assets, assessed the likelihood and severity of each risk, and used a risk register to prioritize remediation efforts. The project demonstrates how organizations identify, analyze, and manage cybersecurity risks to support informed decision-making.

2. Risk Assessment Overview

Asset assessed : bank funds

Evaluation method : risk priority = likelihood x severity

The assessment used a qualitative risk matrix to determine which risks required the most immediate attention.

3. Risk Evaluated

a) financial records leak

Description : a database server containing backup data is publicly accessible.

Likelihood : 3

Severity : 3

Risk score : 9

Priority : Critical

Reason: Exposure of financial records could result in regulatory penalties, reputational damage, and financial loss.

b) compromised user database

Description : customer data is protected with weak encryption

Likelihood : 2

Severity : 3

Risk score : 6

Priority : High

Reason: Weak protection increases the likelihood of unauthorized access to customer information.

c) business email compromise

Description : an employee is tricked into sharing confidential information.

Likelihood : 2

Severity : 2

Risk score : 4

Priority : Medium

Reason: Phishing and social engineering attacks remain common threats in financial institutions.

d) theft

Description : the bank safe is left unloc

Likelihood : 1

Severity : 3

Risk score : 3

Priority : low-medium

Reason: The bank operates in a low-crime area, reducing overall likelihood.

e) supply chain disruption

Description :delivery delays caused by natural disasters.

Likelihood : 1

Severity : 2

Risk score : 2

Priority : low

Reason: while disruptive, the overall impact on operations is limited compared with data-related threats.

4. Key Findings

- Highest risk : financial records leak (score 9) - this risk should receive immediate remediation due to its high likelihood and severe business impact.
- Secondary risks : compromised user database (score 6) and business email compromise (score 4). These risks should be addressed through stronger security controls, employee awareness, and improved data protection mechanisms.

5. Skills demonstrated

- risk assessment
- Risk prioritization
- Risk register development
- Cybersecurity governance
- Asset protection
- Threat analysis
- Security planning
- Risk scoring methodologies
- NIST Cybersecurity Framework Concepts
- Business Impact Analysis.

6. Summary

This project focused on assessing cybersecurity risks affecting a commercial banking environment. I evaluated multiple threats by measuring likelihood and potential business impact, calculated risk scores, and prioritized remediation activities using a risk register. The exercise demonstrated how structured risk Assessment help organizations allocate resources effectively and reduce exposure to critical threats.